IRREDUCIBLE COMPONENTS PULL BACK TO UNIONS OF IRREDUCIBLE COMPONENTS UNDER A FIELD EXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof leans on the standard fact that generalizations lift along flat morphisms; the author invokes it in one clause and the clause is reproduced verbatim.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-inverse-image-irreducible is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Generalizations lift along flat morphisms. Same chapter's toolbox, invoked at proof line 1605 as a standard property of the flat morphism T_K to T. Supplies that every point of T_K generic in an irreducible component of T_K maps to the generic point of T.
- Generic points of irreducible components and specializations. Standard scheme theory, used at proof lines 1608-1610. Supplies the step that a specialization of a point of the inverse image has no escape from T_K.
- Base change of a scheme along a field extension. Standard, the formation of X_K and of the projection p: X_K to X. Supplies the objects whose components are being compared.

Lemma 0.1. *Let K/k be an extension of fields. Let X be a scheme over k. For every irreducible component T of X the inverse image $T_K \subset X_K$ is a union of irreducible components of X_K.*

Proof. Let $T \subset X$ be an irreducible component of X. The morphism $T_K \to T$ is flat, so generalizations lift along $T_K \to T$. Hence every $\xi \in T_K$ which is a generic point of an irreducible component of T_K maps to the generic point η of T. If $\xi' \rightsquigarrow \xi$ is a specialization in X_K then ξ' maps to η since there are no points specializing to η in X. Hence $\xi' \in T_K$ and we conclude that $\xi = \xi'$. In other words ξ is the generic point of an irreducible component of X_K. This means that the irreducible components of T_K are all irreducible components of X_K. □

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.